

RP Pornmongkolsuk

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Summary

A Ph.D. candidate in population genetics with a natural curiosity about human adaptations, a strong foundation in statistical and computational approaches, and a deep passion for engaging non-scientific audiences through multimedia storytelling.

Education

PhD **Duke University**, PhD in Genetics and Genomics – Durham, NC Aug 2021 – present

BA **Brown University**, BA in Biology – Providence, RI Aug 2017 – May 2021

graduate research experience

Alejandro Ochoa Lab, Graduate Researcher – Duke University May 2022 – present

- *Project*: Leveraging ancient DNA in studying human adaptation to vivax malaria in Africa in admixture context.
- Developing a novel, regression-based single-nucleotide imputation approach that improves imputation accuracy on ancient African DNA which are challenging due to (1) low coverage of ancient DNA sequences and (2) low linkage disequilibrium among African populations.
- Benchmarked performance of the low-coverage imputation tool GLIMPSE2 in the imputation of ancient African genomes using simulations (SLiM, Msprime, Tskit).
- Built and performed pipelines for nucleotide imputation of the Duffy antigen receptor locus with custom African reference panel using Minimac4 and Beagle5.4 on ancient African genomes. Performance evaluation of above imputations was done via leave-one-out cross-validation.
- Characterized population structures in ancient and modern African human sequencing data using Principal Component Analyses (PCA), F_{ST} , F -statistics, ADMIXTURE software and the popkin R package.

Amy Goldberg Lab, Rotation Graduate Researcher – Duke University Mar 2022 – Apr 2022

- Developed a pipeline for joint analyses of archaeological (radiocarbon dating) and genetic data (whole genome sequences) in studying the Neolithic Revolution—a wide-scale transition into agricultural society promoting population growth—in East Asian populations.

Craig Lowe Lab, Rotation Graduate Researcher – Duke University Jan 2022 – Feb 2022

- Designed and programmed a genome simulation algorithm as part of the novel genomic toolkit (Gonomics) using Golang.
- The program simulates genetic sequences from populations under the Wright-Fisher model with adjustable (1) population size, (2) number of generations, (3) mutation rate & (4) selection coefficient. The simulation extends beyond the Wright-Fisher model's infinite-site assumption, allowing back mutation and multiallelic polymorphism in the simulated sequences.

Jenny Tung Lab, Rotation Graduate Researcher – Duke University Nov 2021 – Dec 2021

- Built a linear mixed-effect regression model for studying the effects of energy expenditure (3-dimensional motion measured by accelerometers) on gene expression (bulk RNAseq data) in Kalahari meerkats. The model included random effects from kinship coefficients from known pedigrees.

Mohamed Noor Lab, Rotation Graduate Researcher – Duke University Sept 2021 – Oct 2021

- Implemented the GATK best practice pipeline to prepare and analyze sequencing data generated from *Drosophila melanogaster*.

undergraduate research experience

Emilia Huerta-Sanchez lab, Undergraduate Research Assistant – Brown University

June 2020 – May 2021

- Detected signals of adaptive introgression in MUC19, a gene encoding a gel-forming mucin protein, from Denisovans in admixed American populations.
- Assisted in the identification of Native American and European ancestries in the American populations from the 1000 Genomes Project.
- Calculated genetic distances between haplotypes from Native American ancestries in comparison to the Denisovan haplotype using the program Haplostrips.
- Constructed and analyzed the Principal Components (PCA) of genetic distances among the admixed American populations.
- Designed, built, and maintained the Huerta-Sanchez laboratory website on Weebly platform, created illustrations of the research topics page on the lab website.

Laboratory of Animal Cytogenetics and Comparative Genomics (ACCG), Undergraduate Research Assistant – Kasetsart University, Bangkok, Thailand

June 2019 – July 2019

- Conducted and assisted conservation studies of vertebrate species from different zoos in Thailand
- Performed DNA extraction and Polymerase Chain Reaction (PCR) from blood samples collected from the zoos (Bats and Macaques)
- Utilized various molecular biology techniques to study the genetic diversity and structure of Lyle's Flying Foxes (*Pteropus lylei*) in Thailand
- Collected bone marrow tissue from amphibian specimens for Fluorescence in situ Hybridization (FISH) in cytogenetic studies
- Assisted in FISH to examine sex determination patterns in the amphibian species
- Facilitated the Thai-English translations of scientific manuscripts and grant proposals

teaching experience

BIO202: Intro to Genetics and Evolution – Duke University

Fall semesters 2025 & 2026

Graduate Teaching Assistant

- Independently supervised and taught weekly lab sections.
- Assisted students in understanding genetics and evolutionary concepts during office hours and exam reviews.
- Engaged in and lead Canvas discussions on genetics and evolution in the news.
- Graded weekly lab assignments.

Science Day – Creech Road Elementary, Clayton, NC

March 2025

Guest Instructor

- Conducted a hands-on lesson on using Monster Maker game and candy DNA exercise for fifth graders at Creech Road Elementary School.

Intro to Data Analysis – North Carolina School of Science and Mathematics (NCSSM)

January 2025

Instructor

- Instructed an introduction to data analysis using R to the high-school students during their January Term at the North Carolina School of Science and Mathematics (NCSSM). Experience shared on LinkedIn.
- Led a poster design session where students designed posters for presenting their January Term research findings.
- Selected testimonies: * "Goated lesson. Very engaging and helpful, like the flexibility, energy, and analogy explanations."
* "Loved the class! Really helped me get a better grasp on programming with R..."

Bioinformatics Bootcamp – Duke University

Summers 2022-2024

Instructor

- Designed and instructed various bioinformatics lessons for the incoming PhD students in the Genetics and Genomics program: * 2024 lesson: Data Exploration and Cleanup in R. * 2023 lessons: Introduction to Programming Using Python & Data Exploration Using Jupyter Notebook. * 2022 lesson: Introduction to Unix and Command Line.
- Adapted and maintained the Bootcamp website via GitHub.

Genetics – Brown University

Falls 2018 & 2019

Undergraduate Teaching Assistant

- Held office hours for weekly assignments, taught in review sections for exams, graded the weekly assignments.

leadership experience

Committee of Alumni, Marketing, and PR (cAMP) – Duke University

Oct 2024 – May 2026

Co-Chair

- Lead a committee focusing on building the UPGG alumni network on LinkedIn, creating and maintaining positive public image for the students in the program, and engage the student body with the broader scientific community on social media.

Distinguished Lecture Series Committee – Duke University

Oct 2022 – Jan 2025

Co-Chair

- Organized the Distinguished Lecture Series in which we invite distinguished scientists who have made notable contributions to their fields and are great mentors and communicators.

talks, conferences, and workshops

ComSciCon-Triangle Workshop – Durham, NC

January 2026

Participant

- Participated in a Write-A-Thon which involved writing a science news article covering interesting recent scientific finding (in-progress).
- Gave a one-minute, impromptu pop talk about my research (video uploaded on YouTube).
- Engaged with a panel of science writing experts on topics including how to write a science article, use effective visual storytelling, and more. Experience shared on LinkedIn.

American Society of Human Genetics (ASHG) Annual Meeting – Boston, MA

November 2025

Science Talk '25 – Raleigh, NC

April 2025

Participant

- Participated in three-day conference on various topics on science communications.

Alan Alda Center for Communicating Science Workshop – Durham, NC

January 2025

Participant

- Engaged in two-day hands-on science communication workshop using improvisational techniques based on the Alda Method. Experience shared on LinkedIn.

ComSciCon-Triangle Workshop – Chapel Hill, NC

January 2025

Participant

- Participated in a Write-A-Thon which involved writing a science news article covering interesting recent scientific finding (in-progress).
- Gave a one-minute, impromptu pop talk about my research (video uploaded on YouTube).
- Engaged with a panel of science writing experts on topics including how to write a science article, use effective visual storytelling, and more. Experience shared on LinkedIn.

American Society of Human Genetics (ASHG) Annual Meeting – Denver, CO

November 2024

Poster Presentation

- Poster presentation: Evaluation of imputation methods for ancient African DNA.

GradX Talk – Duke University

March 2024

Speaker

- A 5-minute talk titled: How population genetics helps inform us of how we became who we are to general audience.

American Society of Human Genetics (ASHG) Annual Meeting – Washington, D.C.

November 2023

Poster Presentation

- Poster presentation: Single-locus imputation of ancient African DNA using novel regression-based approach.

Communicate Your Science through Short Video Workshop – Duke University

August 2023

Participant

- Wrote, filmed, and edited a science communication short video (YouTube or LinkedIn).

Society for Molecular Biology & Evolution (SMBE) Annual Meeting – Remote Attendance

July 2023

Remote Attendee

AGAR Workshop on Computational Genetics – Remote Attendance

July 2022

Participant

- Hands-on workshop on Snakemake, Msprime, Admixtools & Legofit.

Science at the Intersection of Genetics, Identity, and Health Workshop – Duke University

November 2022

Participant

- Discussed the current topics on science of human genetics and how their implications on identity and health.

Adaptation in Structured Population Workshop – Remote Attendance

June 2023

Participant

scholarships and honors

Associate Membership – Research Triangle Park, NC

2025 – present

Sigma Xi The Scientific Research Honor Society

Royal Thai Government Scholarship – Washington, D.C.

2016 – 2021

Office of Educational Affairs, Royal Thai Embassy

Silver Medal – Aarhus, Denmark

2015

International Biology Olympiad (IBO)

Skills

General: High-performance computing clusters, Graphic design, video editing, public speaking

Languages: Thai, English

Programming: R, Python, Shell, Golang, Java

Software: Plink, Bcftools, GLIMPSE, ADMIXTURE, Minimac, Beagle, Snakemake, Admixtools, SLiM, Msprime, GATK & Haplostrips.

References

Professor John Doe

Professor Jane Smith